

Transforms

- Transforms are a powerful tool for simplifying the analysis of signals and of linear systems.
- Interesting transforms for us:
 - Linearity applies:

$$T[ax + by] = aT[x] + bT[y]$$

- Convolution is replaced by simpler operation:

$$T[x * y] = T[x]T[y]$$

...Transforms

- Most commonly transforms that used in communications engineering are:
 - Laplace transforms (Continuous in Time & Frequency)
 - Continuous Fourier transforms (Continuous in Time)
 - Discrete Fourier transforms (Discrete in Time)
 - Z transforms (Discrete in Time)

THE Z-TRANSFORM

The Z-Transform

- Counterpart of the Laplace transform for discrete-time signals
- Generalization of the Fourier Transform
 - Fourier Transform does not exist for all signals*

- Definition:

$$X(z) = \sum_{n=-\infty}^{\infty} x[n] z^{-n}$$

- Compare to DTFT definition:

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n}$$

- z is a complex variable that can be represented as $z = r e^{j\omega}$
- Substituting $z = e^{j\omega}$ will reduce the z -transform to DTFT

The Z-Transform

تبدیل Z دو طرفه

$$Z\{x[n]\} = \sum_{n=-\infty}^{\infty} x[n]z^{-n} = X(z)$$

$$x[n] \xleftrightarrow{z} X(z)$$

ω

$$X(z) = \sum_{n=0}^{\infty} x[n]z^{-n}$$

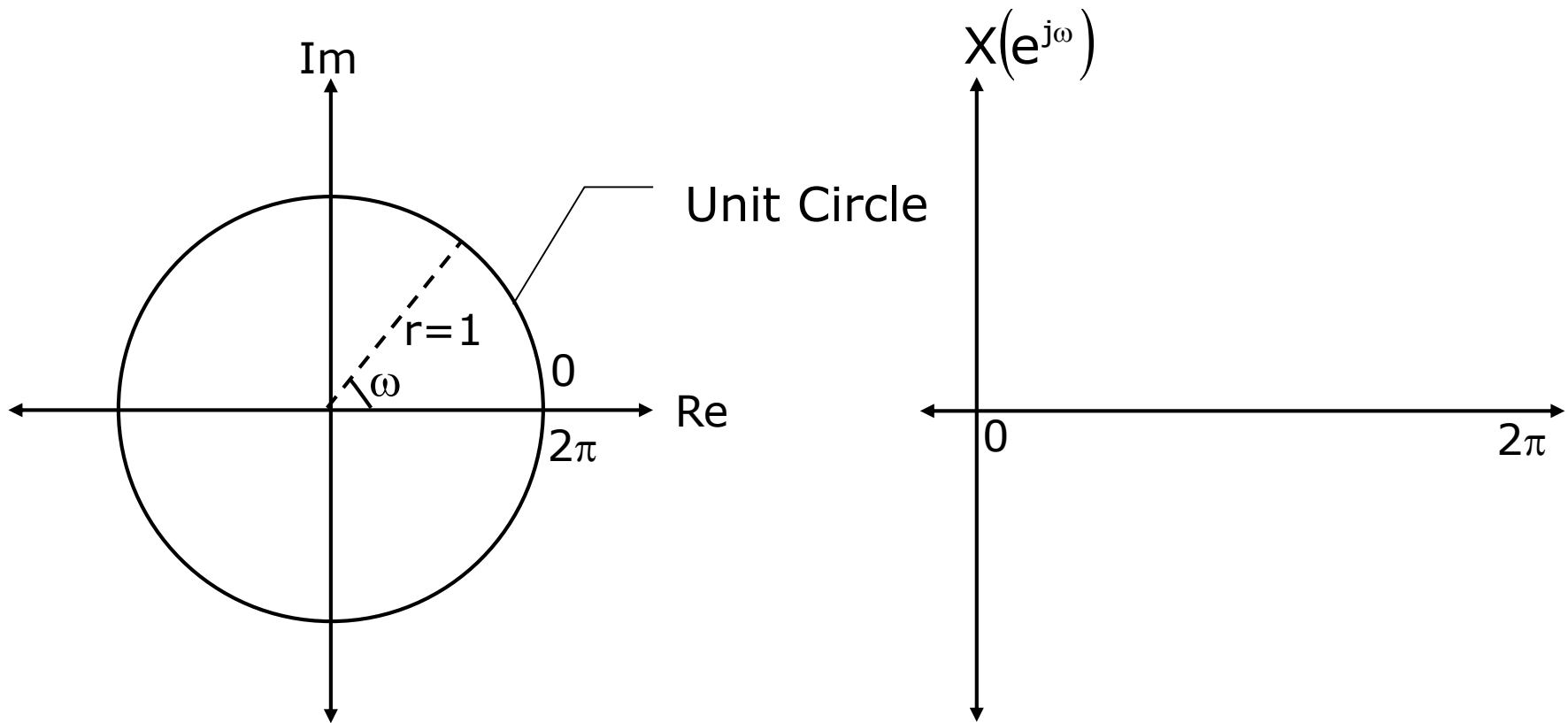
تبدیل Z یک طرفه

$$z = re^{j\omega}$$

r : اندازه : فاز

The z-transform and the DTFT

- Convenient to describe on the complex z-plane
- If we plot $z=e^{j\omega}$ for $\omega=0$ to 2π we get the unit circle



Convergence of the z-Transform

- DTFT does not always converge $X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n}$
Example: $x[n] = a^n u[n]$ for $|a| > 1$ does not have a DTFT

- Complex variable z can be written as $r e^{j\omega}$ so the z-transform

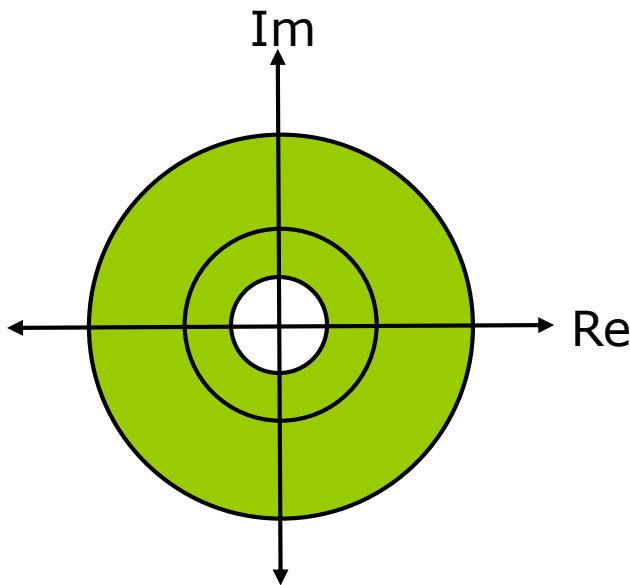
$$X(re^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n](re^{-j\omega})^{-n} = \sum_{n=-\infty}^{\infty} (x[n]r^{-n})e^{-j\omega n}$$

convert to the DTFT of $x[n]$ multiplied with exponential sequence r^{-n}

- For certain choices of r the sum
maybe made finite
$$\sum_{n=-\infty}^{\infty} |x[n]r^{-n}| < \infty$$

Region of Convergence (ROC)

- ROC: The set of values of z for which the z -transform converges
- The region of convergence is made of circles



- **Example:** z -transform converges for values of $0.5 < r < 2$

ROC is shown on the left

In this example the ROC includes the unit circle, so DTFT exists

Example 1: Right-Sided Exponential Sequence

$$x[n] = a^n u[n] \Rightarrow X(z) = \sum_{n=-\infty}^{\infty} a^n u[n] z^{-n} = \sum_{n=0}^{\infty} (az^{-1})^n$$

- For Convergence we require

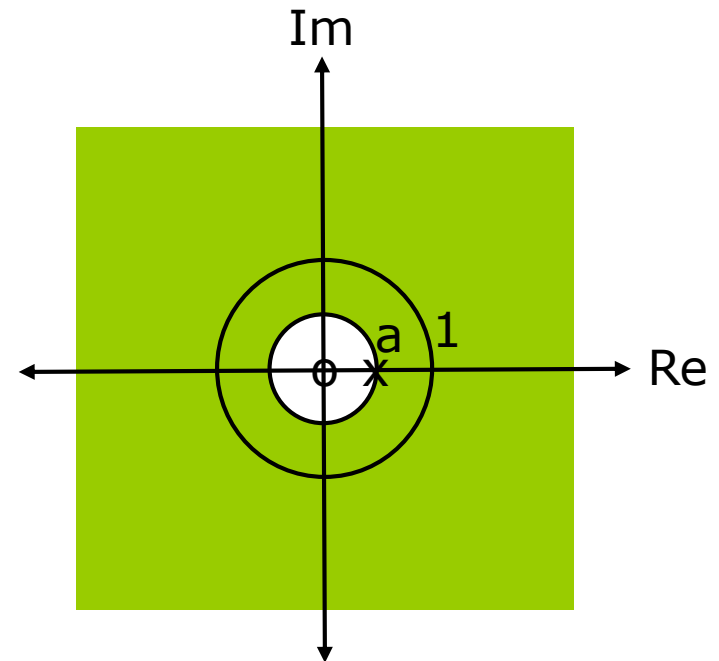
$$\sum_{n=0}^{\infty} |az^{-1}|^n < \infty$$

- Hence the ROC is defined as

$$|az^{-1}|^n < 1 \Rightarrow |z| > |a|$$

- Inside the ROC series converges to

$$X(z) = \sum_{n=0}^{\infty} (az^{-1})^n = \frac{1}{1 - az^{-1}} = \frac{z}{z - a}$$



- Region outside the circle of radius a is the ROC
- Right-sided sequence ROCs extend outside a circle

Example 2: Left-Sided Exponential Sequence

$$x[n] = -a^n u[-n-1]$$

(دنبالہ چپگرا)

$$\begin{aligned} X(z) &= -\sum_{n=-\infty}^{\infty} a^n u[-n-1] z^{-n} = -\sum_{n=-\infty}^{-1} a^n z^{-n} \\ &= -\sum_{n=1}^{\infty} (a^{-1}z)^n = 1 - \sum_{n=0}^{\infty} (a^{-1}z)^n \end{aligned}$$

ROC :

$$\sum_{n=0}^{\infty} |a^{-1}z|^n < \infty \Rightarrow |a^{-1}z| < 1 \Rightarrow |z| < |a|$$

$$X(z) = 1 - \frac{1}{1 - a^{-1}z} = \frac{1}{1 - az^{-1}} = \frac{z}{z - a}$$

Example 3: Two-Sided Exponential Sequence

$$x[n] = \left(-\frac{1}{3}\right)^n u[n] - \left(\frac{1}{2}\right)^n u[-n-1]$$

$$\sum_{n=0}^{\infty} \left(-\frac{1}{3} z^{-1}\right)^n = \frac{\left(-\frac{1}{3} z^{-1}\right)^0 - \left(-\frac{1}{3} z^{-1}\right)^{\infty}}{1 + \frac{1}{3} z^{-1}} = \frac{1}{1 + \frac{1}{3} z^{-1}}$$

$$\sum_{n=-\infty}^{-1} \left(\frac{1}{2} z^{-1}\right)^n = \frac{\left(\frac{1}{2} z^{-1}\right)^{-\infty} - \left(\frac{1}{2} z^{-1}\right)^0}{1 - \frac{1}{2} z^{-1}} = \frac{-1}{1 - \frac{1}{2} z^{-1}}$$

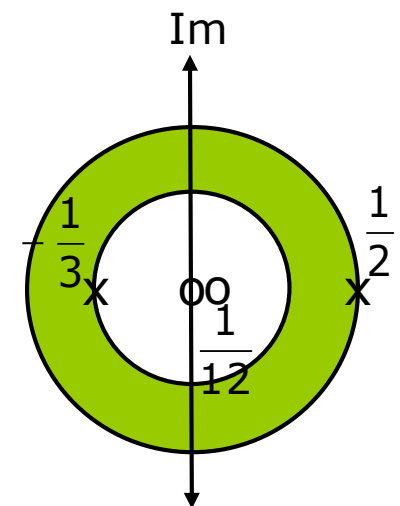
$$X(z) = \frac{1}{1 + \frac{1}{3} z^{-1}} + \frac{1}{1 - \frac{1}{2} z^{-1}} = \frac{2z \left(z - \frac{1}{12}\right)}{\left(z + \frac{1}{3}\right) \left(z - \frac{1}{2}\right)}$$

$$\text{ROC: } \left| -\frac{1}{3} z^{-1} \right| < 1$$

$$\frac{1}{3} < |z|$$

$$\text{ROC: } \left| \frac{1}{2} z^{-1} \right| > 1$$

$$\frac{1}{2} > |z|$$



Example 4: Finite Length Sequence

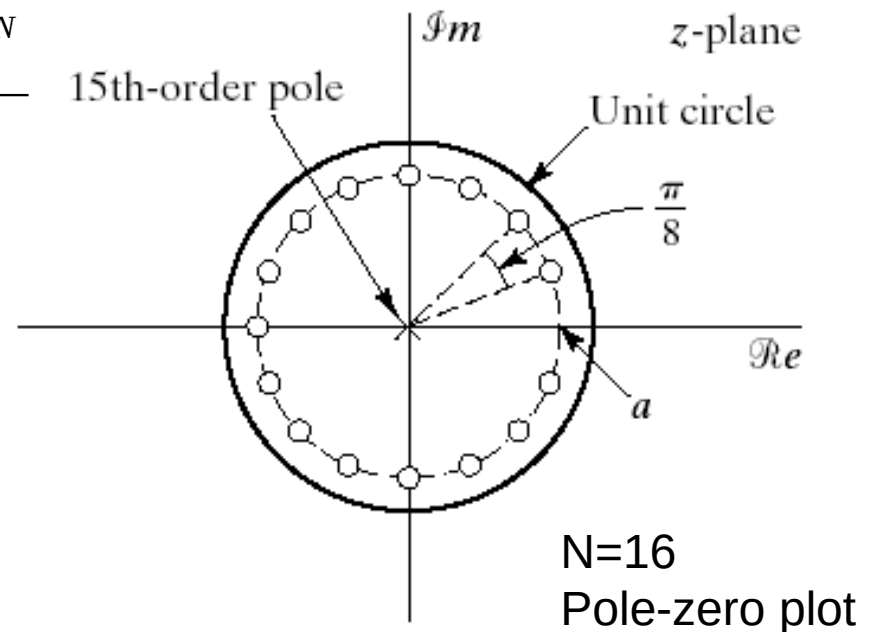
$$x[n] = a^n (u[n] - u[n - N])$$

$$x[n] = \begin{cases} a^n & 0 \leq n \leq N - 1 \\ 0 & \text{otherwise} \end{cases}$$

$$\begin{aligned} X(z) &= \sum_{n=0}^{N-1} a^n z^{-n} = \sum_{n=0}^{N-1} (az^{-1})^n = \frac{1 - (az^{-1})^N}{1 - az^{-1}} \\ &= \frac{1}{z^{N-1}} \cdot \frac{z^N - a^N}{z - a} \end{aligned}$$

ROC :

$$\sum_{n=0}^{N-1} |az^{-1}|^n < \infty \Rightarrow |az^{-1}| < \infty \Rightarrow z \neq 0$$



Some common Z-transform pairs

Sequence	Transform	ROC
1. $\delta[n]$	1	All z
2. $u[n]$	$\frac{1}{1 - z^{-1}}$	$ z > 1$
3. $-u[-n - 1]$	$\frac{1}{1 - z^{-1}}$	$ z < 1$
4. $\delta[n - m]$	z^{-m}	All z except 0 (if $m > 0$) or ∞ (if $m < 0$)
5. $a^n u[n]$	$\frac{1}{1 - az^{-1}}$	$ z > a $
6. $-a^n u[-n - 1]$	$\frac{1}{1 - az^{-1}}$	$ z < a $
7. $na^n u[n]$	$\frac{az^{-1}}{(1 - az^{-1})^2}$	$ z > a $
8. $-na^n u[-n - 1]$	$\frac{az^{-1}}{(1 - az^{-1})^2}$	$ z < a $
9. $[\cos \omega_0 n] u[n]$	$\frac{1 - [\cos \omega_0] z^{-1}}{1 - [2 \cos \omega_0] z^{-1} + z^{-2}}$	$ z > 1$
10. $[\sin \omega_0 n] u[n]$	$\frac{[\sin \omega_0] z^{-1}}{1 - [2 \cos \omega_0] z^{-1} + z^{-2}}$	$ z > 1$
11. $[r^n \cos \omega_0 n] u[n]$	$\frac{1 - [r \cos \omega_0] z^{-1}}{1 - [2r \cos \omega_0] z^{-1} + r^2 z^{-2}}$	$ z > r$
12. $[r^n \sin \omega_0 n] u[n]$	$\frac{[r \sin \omega_0] z^{-1}}{1 - [2r \cos \omega_0] z^{-1} + r^2 z^{-2}}$	$ z > r$
13. $\begin{cases} a^n, & 0 \leq n \leq N - 1, \\ 0, & \text{otherwise} \end{cases}$	$\frac{1 - a^N z^{-N}}{1 - az^{-1}}$	$ z > 0$

Some common Z-transform pairs

SEQUENCE	TRANSFORM	ROC
$\delta[n]$	1	ALL z
$u[n]$	$\frac{1}{1 - z^{-1}}$	$ z > 1$
$-u[-n-1]$	$\frac{1}{1 - z^{-1}}$	$ z < 1$
$\delta[n-m]$	z^{-m}	All z except 0 (if $m > 0$) or ∞ (if $m < 0$)

Some common Z-transform pairs

$$a^n u[n] \quad \longleftrightarrow \quad \frac{1}{1 - az^{-1}} \quad \text{ROC : } |z| > |a|$$

$$a^n u[-n-1] \quad \longleftrightarrow \quad \frac{1}{1 - az^{-1}} \quad \text{ROC : } |z| < |a|$$

$$na^n u[n] \quad \longleftrightarrow \quad \frac{az^{-1}}{(1 - az^{-1})^2} \quad \text{ROC : } |z| > |a|$$

$$-na^n u[-n-1] \quad \longleftrightarrow \quad \frac{az^{-1}}{(1 - az^{-1})^2} \quad \text{ROC : } |z| < |a|$$

$$[\cos \omega_0 n] u[n] \quad \longleftrightarrow \quad \frac{1 - [\cos \omega_0] z^{-1}}{1 - [2 \cos \omega_0] z^{-1} + z^{-2}} \quad \text{ROC : } |z| > 1$$

Some common Z-transform pairs

$$[\sin \omega_0 n]u[n] \xleftrightarrow{z} \frac{[\sin \omega_0]z^{-1}}{1 - [2 \cos \omega_0]z^{-1} + z^{-2}} \quad ROC: |z| > 1$$

$$[r^n \cos \omega_0 n]u[n] \xleftrightarrow{z} \frac{1 - [r \cos \omega_0]z^{-1}}{1 - [2r \cos \omega_0]z^{-1} + r^2 z^{-2}} \quad ROC: |z| > r$$

$$[r^n \sin \omega_0 n]u[n] \xleftrightarrow{z} \frac{[r \sin \omega_0]z^{-1}}{1 - [2r \cos \omega_0]z^{-1} + r^2 z^{-2}} \quad ROC: |z| > r$$

$$\begin{cases} a^n & 0 \leq n \leq N-1 \\ 0 & \text{otherwise} \end{cases} \xleftrightarrow{z} \frac{1 - a^N z^{-N}}{1 - az^{-1}} \quad ROC: |z| > 0$$

Properties of The ROC of Z-Transform

- The ROC is a ring or disk centered at the origin
- DTFT exists if and only if the ROC includes the unit circle
- The ROC cannot contain any poles
- The ROC for finite-length sequence is the entire z -plane except possibly $z=0$ and $z=\infty$
- The ROC for a right-handed sequence extends outward from the outermost pole possibly including $z= \infty$
- The ROC for a left-handed sequence extends inward from the innermost pole possibly including $z=0$
- The ROC of a two-sided sequence is a ring bounded by poles
- The ROC must be a connected region
- A z -transform does not uniquely determine a sequence without specifying the ROC

Example

EXAMPLE Find the z -transform of $x(n] = (\frac{1}{2})^n u(n) - 2^n u(-n - 1)$, and find another signal that has the same z -transform but a different region of convergence.

Here we have a sum of two sequences. Therefore, we may find the z -transform of each sequence separately and add them together.

the z -transform of $x_1(n] = (\frac{1}{2})^n u(n)$ is
$$X_1(z) = \frac{1}{1 - \frac{1}{2}z^{-1}} \quad |z| > \frac{1}{2}$$

the z -transform of $x_2(n] = -2^n u(-n - 1)$ is
$$X_2(z) = \frac{1}{1 - 2z^{-1}} \quad |z| < 2$$

Therefore, the z -transform of $x(n] = x_1(n] + x_2(n]$ is

$$X(z) = \frac{1}{1 - \frac{1}{2}z^{-1}} + \frac{1}{1 - 2z^{-1}} = \frac{2 - \frac{5}{2}z^{-1}}{(1 - \frac{1}{2}z^{-1})(1 - 2z^{-1})}$$

with a region of convergence $\frac{1}{2} < |z| < 2$, which is the set of all points that are in the ROC of both $X_1(z)$ and $X_2(z)$.

Example

To find another sequence that has the same z -transform, note that because $X(z)$ is a sum of two z -transforms,

$$X(z) = \frac{1}{1 - \frac{1}{2}z^{-1}} + \frac{1}{1 - 2z^{-1}}$$

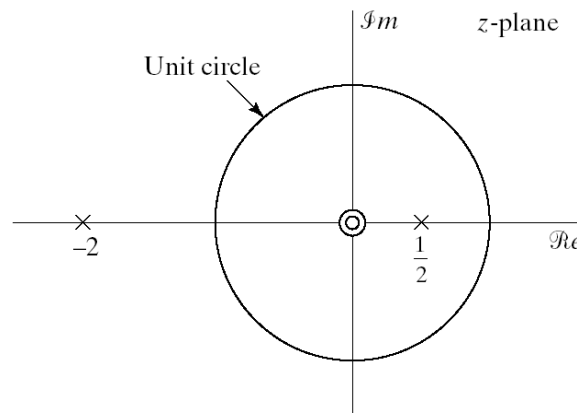
each term corresponds to the z -transform of either a right-sided or a left-sided sequence, depending upon the region of convergence. Therefore, choosing the right-sided sequences for both terms, it follows that

$$x^1(n) = \left(\frac{1}{2}\right)^n u(n) + 2^n u(n)$$

has the same z -transform as $x(n)$, except that the region of convergence is $|z| > 2$.

Stability, Causality, and the ROC

- Consider a system with impulse response $h[n]$
- The z-transform $H(z)$ and the pole-zero plot shown below
- Without any other information $h[n]$ is not uniquely determined
 $|z| > 2$ or $|z| < 1/2$ or $1/2 < |z| < 2$
- If system stable ROC must include unit-circle: $1/2 < |z| < 2$
- If system is causal must be right sided: $|z| > 2$



Z-Transform Properties: Linearity

- Notation

$$x[n] \xleftrightarrow{z} X(z) \quad \text{ROC} = R_x$$

- Linearity

$$ax_1[n] + bx_2[n] \xleftrightarrow{z} aX_1(z) + bX_2(z) \quad \text{ROC} = R_{x_1} \cap R_{x_2}$$

- Note that the ROC of combined sequence may be larger than either ROC
- This would happen if some pole/zero cancellation occurs

- **Example:**

$$x[n] = a^n u[n] - a^n u[n - N]$$

- *Both sequences are right-sided*
- *Both sequences have a pole $z=a$*
- *Both have a ROC defined as $|z| > |a|$*
- *In the combined sequence the pole at $z=a$ cancels with a zero at $z=a$*
- *The combined ROC is the entire z plane except $z=0$*

Z-Transform Properties: Time Shifting

$$x[n - n_0] \xleftrightarrow{z} z^{-n_0} X(z) \quad \text{ROC} = R_x$$

- Here n_0 is an integer
 - If positive the sequence is shifted right
 - If negative the sequence is shifted left
- The ROC can change
 - The new term may add or remove poles at $z=0$ or $z=\infty$
- Example

$$X(z) = z^{-1} \left(\frac{1}{1 - \frac{1}{4} z^{-1}} \right) \quad |z| > \frac{1}{4}$$

$$x[n] = \left(\frac{1}{4} \right)^{n-1} u[n-1]$$

Z-Transform Properties: Multiplication by Exponential

$$z_0^n x[n] \xleftrightarrow{z} X(z/z_0) \quad \text{ROC} = |z_0| R_x$$

- ROC is scaled by $|z_0|$
- All pole/zero locations are scaled
- If z_0 is a positive real number: z-plane shrinks or expands
- If z_0 is a complex number with unit magnitude it rotates
- **Example:** We know the z-transform pair

$$u[n] \xleftrightarrow{z} \frac{1}{1 - z^{-1}} \quad \text{ROC} : |z| > 1$$

- Let's find the z-transform of

$$x[n] = r^n \cos(\omega_0 n) u[n] = \frac{1}{2} (e^{j\omega_0})^n u[n] + \frac{1}{2} (e^{-j\omega_0})^n u[n]$$

$$X(z) = \frac{1/2}{1 - re^{j\omega_0} z^{-1}} + \frac{1/2}{1 - re^{-j\omega_0} z^{-1}} \quad |z| > r$$

Z-Transform Properties: Differentiation

$$nx[n] \xleftrightarrow{z} -z \frac{dX(z)}{dz} \quad \text{ROC} = R_x$$

- **Example:** We want the inverse z-transform of

$$X(z) = \log(1 + az^{-1}) \quad |z| > |a|$$

- Let's differentiate to obtain rational expression

$$\frac{dX(z)}{dz} = \frac{-az^{-2}}{1 + az^{-1}} \Rightarrow -z \frac{dX(z)}{dz} = az^{-1} \frac{1}{1 + az^{-1}}$$

- Making use of z-transform properties and ROC

$$nx[n] = a(-a)^{n-1}u[n-1]$$

$$x[n] = (-1)^{n-1} \frac{a^n}{n} u[n-1]$$

Z-Transform Properties: Conjugation

$$x^*[n] \xleftrightarrow{Z} X^*(z^*) \quad \text{ROC} = R_x$$

$$X(z) = \sum_{n=-\infty}^{\infty} x[n] z^{-n}$$

$$X^*(z) = \left(\sum_{n=-\infty}^{\infty} x[n] z^{-n} \right)^* = \sum_{n=-\infty}^{\infty} x^*[n] z^n$$

$$X^*(z^*) = \sum_{n=-\infty}^{\infty} x^*[n] (z^n)^* = \sum_{n=-\infty}^{\infty} x^*[n] z^{-n} = Z \{x^*[n]\}$$

Z-Transform Properties: Time Reversal

$$x[-n] \xleftrightarrow{z} X(1/z) \quad \text{ROC} = \frac{1}{R_x}$$

- ROC is inverted
- **Example:** $x[n] = a^{-n}u[-n]$

- Time reversed version of $u[n]$

$$X(z) = \frac{1}{1 - az} = \frac{-a^{-1}z^{-1}}{1 - a^{-1}z^{-1}} \quad |z| < |a^{-1}|$$

Z-Transform Properties: Convolution

$$x_1[n] * x_2[n] \xrightarrow{z} X_1(z)X_2(z) \quad \text{ROC} : R_{x_1} \cap R_{x_2}$$

- Convolution in time domain is multiplication in z-domain
- **Example:** Let's calculate the convolution of

$$x_1[n] = a^n u[n] \quad \text{and} \quad x_2[n] = u[n]$$

$$X_1(z) = \frac{1}{1 - az^{-1}} \quad \text{ROC} : |z| > |a| \quad X_2(z) = \frac{1}{1 - z^{-1}} \quad \text{ROC} : |z| > 1$$

- Multiplications of z-transforms is

$$Y(z) = X_1(z)X_2(z) = \frac{1}{(1 - az^{-1})(1 - z^{-1})}$$

- ROC: if $|a| < 1$ ROC is $|z| > 1$ if $|a| > 1$ ROC is $|z| > |a|$
- Partial fractional expansion of $Y(z)$

$$Y(z) = \frac{1}{1-a} \left(\frac{1}{1-z^{-1}} - \frac{a}{1-az^{-1}} \right) \quad \text{assume ROC} : |z| > 1 \quad y[n] = \frac{1}{1-a} (u[n] - a^{n+1}u[n])$$

Some Z-transform properties

Section Reference	Sequence	Transform	ROC
	$x[n]$	$X(z)$	R_x
	$x_1[n]$	$X_1(z)$	R_{x_1}
	$x_2[n]$	$X_2(z)$	R_{x_2}
3.4.1	$ax_1[n] + bx_2[n]$	$aX_1(z) + bX_2(z)$	Contains $R_{x_1} \cap R_{x_2}$
3.4.2	$x[n - n_0]$	$z^{-n_0} X(z)$	R_x , except for the possible addition or deletion of the origin or ∞
3.4.3	$z_0^n x[n]$	$X(z/z_0)$	$ z_0 R_x$
3.4.4	$nx[n]$	$-z \frac{dX(z)}{dz}$	R_x , except for the possible addition or deletion of the origin or ∞
3.4.5	$x^*[n]$	$X^*(z^*)$	R_x
	$\text{Re}\{x[n]\}$	$\frac{1}{2}[X(z) + X^*(z^*)]$	Contains R_x
	$\text{Im}\{x[n]\}$	$\frac{1}{2j}[X(z) - X^*(z^*)]$	Contains R_x
3.4.6	$x^*[-n]$	$X^*(1/z^*)$	$1/R_x$
3.4.7	$x_1[n] * x_2[n]$	$X_1(z)X_2(z)$	Contains $R_{x_1} \cap R_{x_2}$
3.4.8	Initial-value theorem:		
	$x[n] = 0, \quad n < 0$	$\lim_{z \rightarrow \infty} X(z) = x[0]$	

The Inverse Z-Transform

- Formal inverse z-transform is based on a Cauchy integral
- Less formal ways sufficient most of the time
 - *Inspection method*
 - *Partial fraction expansion*
 - *Power series expansion*

- Inspection Method

Make use of known z-transform pairs such as

$$a^n u[n] \xleftrightarrow{z} \frac{1}{1 - az^{-1}} \quad |z| > |a|$$

Example: The inverse z-transform of

$$X(z) = \frac{1}{1 - \frac{1}{2}z^{-1}} \quad |z| > \frac{1}{2} \quad \rightarrow \quad x[n] = \left(\frac{1}{2}\right)^n u[n]$$

Inverse Z-Transform by Partial Fraction Expansion

- Assume that a given z-transform can be expressed as

$$X(z) = \frac{\sum_{k=0}^M b_k z^{-k}}{\sum_{k=0}^N a_k z^{-k}}$$

- Apply partial fractional expansion

$$X(z) = \sum_{r=0}^{M-N} B_r z^{-r} + \sum_{k=1, k \neq i}^N \frac{A_k}{1 - d_k z^{-1}} + \sum_{m=1}^s \frac{C_m}{(1 - d_i z^{-1})^m}$$

- First term exist only if $M > N$
 - B_r is obtained by long division
- Second term represents all first order poles
- Third term represents an order s pole
 - There will be a similar term for every high-order pole
- Each term can be inverse transformed by inspection

Inverse Z-Transform by Partial Fraction Expansion

$$X(z) = \sum_{r=0}^{M-N} B_r z^{-r} + \sum_{k=1, k \neq i}^N \frac{A_k}{1 - d_k z^{-1}} + \sum_{m=1}^s \frac{C_m}{(1 - d_i z^{-1})^m}$$

- Coefficients are given as

$$A_k = (1 - d_k z^{-1}) X(z) \Big|_{z=d_k}$$

$$C_m = \frac{1}{(s-m)! (-d_i)^{s-m}} \left\{ \frac{d^{s-m}}{dw^{s-m}} [(1 - d_i w)^s X(w^{-1})] \right\}_{w=d_i^{-1}}$$

- Easier to understand with examples

Example 5: 2nd Order Z-Transform

$$X(z) = \frac{1}{\left(1 - \frac{1}{4}z^{-1}\right)\left(1 - \frac{1}{2}z^{-1}\right)} \quad \text{ROC: } |z| > \frac{1}{2}$$

$$X(z) = \frac{A_1}{\left(1 - \frac{1}{4}z^{-1}\right)} + \frac{A_2}{\left(1 - \frac{1}{2}z^{-1}\right)}$$

$$A_1 = \left(1 - \frac{1}{4}z^{-1}\right)X(z)\Big|_{z=\frac{1}{4}} = \frac{1}{\left(1 - \frac{1}{2}\left(\frac{1}{4}\right)^{-1}\right)} = -1$$

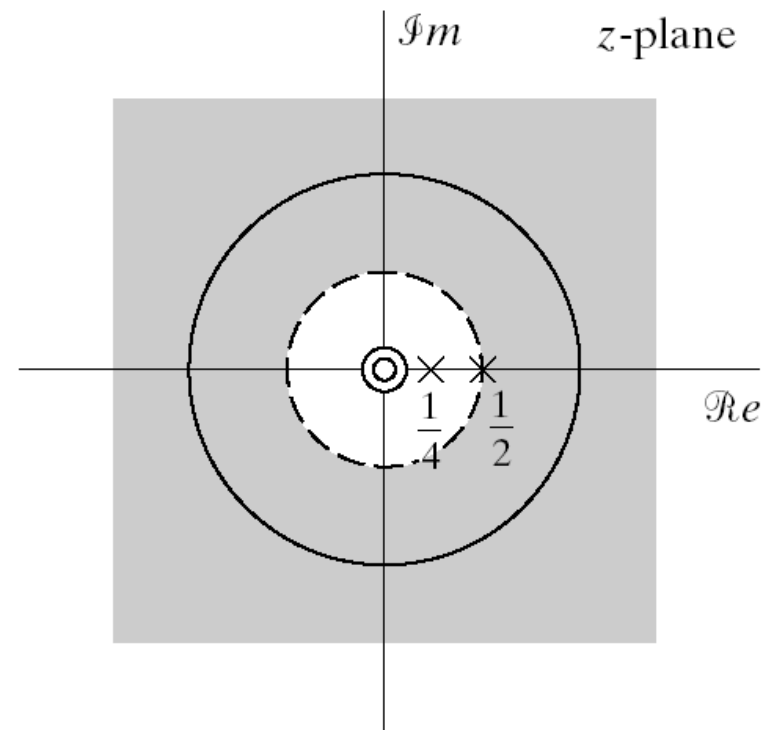
$$A_2 = \left(1 - \frac{1}{2}z^{-1}\right)X(z)\Big|_{z=\frac{1}{2}} = \frac{1}{\left(1 - \frac{1}{4}\left(\frac{1}{2}\right)^{-1}\right)} = 2$$

Example 5 Continued

$$X(z) = \frac{-1}{\left(1 - \frac{1}{4}z^{-1}\right)} + \frac{2}{\left(1 - \frac{1}{2}z^{-1}\right)} \quad |z| > \frac{1}{2}$$

- ROC extends to infinity
 - Indicates right sided sequence

$$x[n] = 2\left(\frac{1}{2}\right)^n u[n] - \left(\frac{1}{4}\right)^n u[n]$$



Example 6

$$X(z) = \frac{1 + 2z^{-1} + z^{-2}}{1 - \frac{3}{2}z^{-1} + \frac{1}{2}z^{-2}} = \frac{(1 + z^{-1})^2}{\left(1 - \frac{1}{2}z^{-1}\right)(1 - z^{-1})} \quad |z| > 1$$

- Long division to obtain B_0

$$\begin{array}{r} \frac{1}{2}z^{-2} - \frac{3}{2}z^{-1} + 1 \overline{) \frac{2}{z^{-2} + 2z^{-1} + 1}} \\ \underline{z^{-2} - 3z^{-1} + 2} \\ 5z^{-1} - 1 \end{array}$$

$$X(z) = 2 + \frac{-1 + 5z^{-1}}{\left(1 - \frac{1}{2}z^{-1}\right)(1 - z^{-1})}$$

$$X(z) = 2 + \frac{A_1}{1 - \frac{1}{2}z^{-1}} + \frac{A_2}{1 - z^{-1}}$$

$$A_1 = \left(1 - \frac{1}{2}z^{-1}\right)X(z) \Big|_{z=\frac{1}{2}} = -9$$

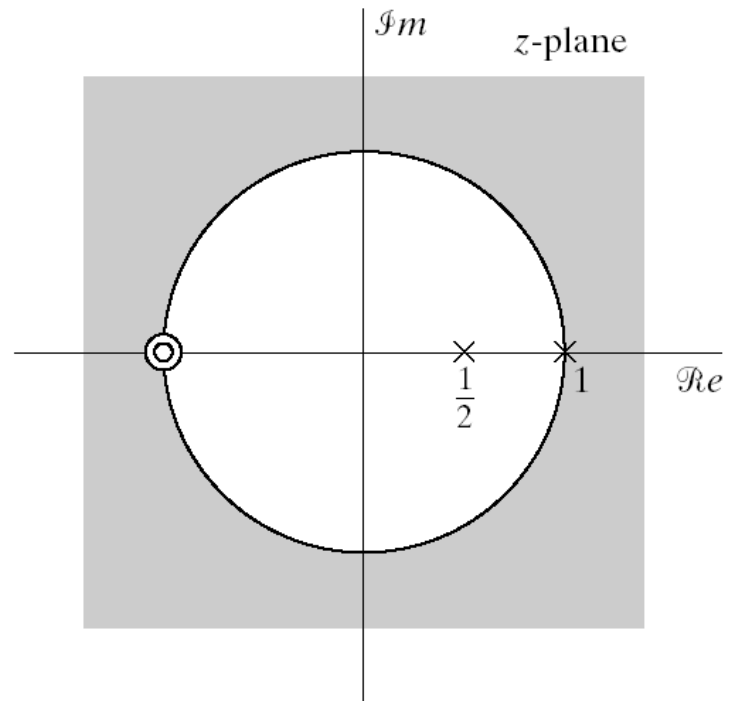
$$A_2 = (1 - z^{-1})X(z) \Big|_{z=1} = 8$$

Example 5 Continued

$$X(z) = 2 - \frac{9}{1 - \frac{1}{2}z^{-1}} + \frac{8}{1 - z^{-1}} \quad |z| > 1$$

- ROC extends to infinity
 - Indicates right-sided sequence

$$x[n] = 2\delta[n] - 9\left(\frac{1}{2}\right)^n u[n] - 8u[n]$$



Inverse Z-Transform by Power Series Expansion

- The z-transform is power series

$$X(z) = \sum_{n=-\infty}^{\infty} x[n] z^{-n}$$

- In expanded form

$$X(z) = \cdots + x[-2]z^2 + x[-1]z^1 + x[0] + x[1]z^{-1} + x[2]z^{-2} + \cdots$$

- Z-transforms of this form can generally be inversed easily
- Especially useful for finite-length series

Example 6

$$\begin{aligned} X(z) &= z^2 \left(1 - \frac{1}{2} z^{-1} \right) (1 + z^{-1}) (1 - z^{-1}) \\ &= z^2 - \frac{1}{2} z - 1 + \frac{1}{2} z^{-1} \end{aligned}$$

$$x[n] = \delta[n+2] - \frac{1}{2} \delta[n+1] - \delta[n] + \frac{1}{2} \delta[n-1]$$

$$x[n] = \begin{cases} 1 & n = -2 \\ -\frac{1}{2} & n = -1 \\ -1 & n = 0 \\ \frac{1}{2} & n = 1 \\ 0 & n = 2 \end{cases}$$